

Sarlink® TPE ML-1155B BLK (PRELIMINARY DATA)

Teknor Apex Company - Thermoplastic Elastomer

Monday, September 18, 2023

General Information

Product Description

Sarlink ML-1100 is a general purpose thermoplastic elastomer series, available in NAT and BLK designed for automotive interior applications. Sarlink ML-1155B BLK is a low hardness, high density, filled grade suitable for injection molding.

General

Generic Name	• Thermoplastic Elastomer (TPE)		
Material Status	• Preliminary Data		
Availability	• Africa & Middle East • Asia Pacific	• Europe • Latin America	• North America
Features	• Chemical Resistant • Filled • Good Adhesion • Good Flexibility	• Good Moldability • Good Tear Strength • Good Toughness • High Density	• High Specific Gravity • Low Flow • Low Hardness • Resilient
Uses	• Automotive Applications • Automotive Interior Parts • Flexible Grips	• General Purpose • Grommets • Knobs	• Rubber Replacement • Soft Touch Applications
RoHS Compliance	• RoHS Compliant		
Appearance	• Black		
Forms	• Pellets		
Processing Method	• Injection Molding		

ASTM & ISO Properties ¹

Physical	Nominal Value	Unit	Test Method
Density	1.18	g/cm ³	ISO 1183
Melt Mass-Flow Rate (MFR) (190°C/2.16 kg)	1.0	g/10 min	ASTM D1238
Elastomers	Nominal Value	Unit	Test Method
Tensile Stress ²			ISO 37
Across Flow : 100% Strain	1.26	MPa	
Flow : 100% Strain	1.50	MPa	
Tensile Stress ²			ISO 37
Across Flow : Break	6.50	MPa	
Flow : Break	5.30	MPa	
Tensile Elongation ²			ISO 37
Across Flow : Break	880	%	
Flow : Break	830	%	
Tear Strength ³			ISO 34-1
Across Flow	20.0	kN/m	
Flow	22.0	kN/m	

Elastomers	Nominal Value	Unit	Test Method
Compression Set ⁴			ISO 815
23°C, 72 hr	14	%	
70°C, 22 hr	32	%	
100°C, 22 hr	69	%	
120°C, 22 hr	78	%	
Hardness	Nominal Value	Unit	Test Method
Shore Hardness (Shore A, Instant, Injection Molded)	52		ISO 868
Aging	Nominal Value	Unit	Test Method
Change in Tensile Strength in Air ⁵			ISO 188
Across Flow : 120°C, 1000 hr	15	%	
Flow : 120°C, 1000 hr	26	%	
Across Flow : 100% Strain 120°C, 1000 hr	14	%	
Flow : 100% Strain 120°C, 1000 hr	13	%	
Change in Tensile Strain at Break in Air ⁵			ISO 188
Across Flow : 120°C, 1000 hr	-0.10	%	
Flow : 120°C, 1000 hr	16	%	
Change in Shore Hardness in Air ⁶			ISO 188
Shore A, 120°C, 1000 hr	4.4		

Legal Statement

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Processing Information

Injection	Nominal Value	Unit
Rear Temperature	182 to 204	°C
Middle Temperature	193 to 227	°C
Front Temperature	204 to 238	°C
Nozzle Temperature	204 to 238	°C
Processing (Melt) Temp	204 to 238	°C
Mold Temperature	21 to 38	°C
Injection Pressure	1.38 to 6.89	MPa
Injection Rate	Moderate-Fast	
Back Pressure	0.172 to 0.345	MPa
Screw Speed	50 to 100	rpm
Cushion	3.81 to 25.4	mm

Injection Notes

Drying is not necessary. However, if moisture is a problem, dry the pellets for 2 to 4 hours at 150°F (65°C).

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Notes

¹ Typical properties: these are not to be construed as specifications.
² Type 2, 500 mm/min
³ Method Ba, Angle (Unnicked), 500 mm/min
⁴ Type A
⁵ Type 2
⁶ Instant